

Spinless crystalline space groups

Crystalline: $s_{\text{crys}} = \omega_{\text{crys}} = 0$. Internal: $s = w_1(V)$, $\omega = w_2(V) + w_1(V)^2$. Full infinite affine symmetry is retained. An abstract upper-extension group does not specify an actual marked upper relation. “Native denotes retained native lifts or their exact reconstruction recipes; the report does not rerun bar operations.

SG	$p + ip$	MC	CF	Boson	Full group / family	Free index	H0 order	Evidence
1	$\mathbb{Z}^3$	$\mathbb{Z}_2^3$	$\mathbb{Z}_2$	0	$\mathbb{Z}^3 \oplus \mathbb{Z}_2^4$	1	–	Native marked
2	$\mathbb{Z}^3$	0	$\mathbb{Z}_2^4$	$\mathbb{Z}_2$	$\mathbb{Z}^3 \oplus \mathbb{Z}_2^5$	8	–	Native marked
3	$\mathbb{Z}$	$\mathbb{Z}_2^3$	$\mathbb{Z}_2^4$	$\mathbb{Z}_2$	$\mathbb{Z} \oplus \mathbb{Z}_2^8$	2	2	Native marked
4	$\mathbb{Z}$	$\mathbb{Z}_2^3$	$\mathbb{Z}_2$	0	$\mathbb{Z} \oplus \mathbb{Z}_2^4$	1	–	Native marked
5	$\mathbb{Z}$	$\mathbb{Z}_2^2$	$\mathbb{Z}_2^2$	$\mathbb{Z}_2$	$\mathbb{Z} \oplus \mathbb{Z}_2^5$	1	2	Native marked
6	$\mathbb{Z} \oplus \mathbb{Z}_2$	$\mathbb{Z}_2^6$	$\mathbb{Z}_2^6$	$\mathbb{Z}_2^3$	$\mathbb{Z} \oplus \mathbb{Z}_2^9 \oplus \mathbb{Z}_8 \oplus \mathbb{Z}_{16}$	1	–	Abstract only
7	$\mathbb{Z} \oplus \mathbb{Z}_2$	$\mathbb{Z}_2^3$	$\mathbb{Z}_2$	0	$\mathbb{Z} \oplus \mathbb{Z}_2^3$	1	–	Native marked
8	$\mathbb{Z} \oplus \mathbb{Z}_2$	$\mathbb{Z}_2^4$	$\mathbb{Z}_2^3$	$\mathbb{Z}_2^2$	$\mathbb{Z} \oplus \mathbb{Z}_2^6 \oplus \mathbb{Z}_{16}$	1	–	Abstract only
9	$\mathbb{Z} \oplus \mathbb{Z}_2$	$\mathbb{Z}_2^2$	$\mathbb{Z}_2$	0	$\mathbb{Z} \oplus \mathbb{Z}_2^4$	1	–	Native marked
10	$\mathbb{Z}$	$\mathbb{Z}_2$	$\mathbb{Z}_2^8$	$\mathbb{Z}_2^7$	$\mathbb{Z} \oplus \mathbb{Z}_2^{11} \oplus \mathbb{Z}_4 \oplus \mathbb{Z}_8$	2	–	Native marked
11	$\mathbb{Z}$	$\mathbb{Z}_2^3$	$\mathbb{Z}_2^5$	$\mathbb{Z}_2^2$	$\mathbb{Z} \oplus \mathbb{Z}_2^7 \oplus \mathbb{Z}_8$	1	–	Native marked
12	$\mathbb{Z}$	$\mathbb{Z}_2$	$\mathbb{Z}_2^5$	$\mathbb{Z}_2^4$	$\mathbb{Z} \oplus \mathbb{Z}_2^7 \oplus \mathbb{Z}_8$	2	–	Native marked
13	$\mathbb{Z}$	$\mathbb{Z}_2^2$	$\mathbb{Z}_2^4$	$\mathbb{Z}_2$	$\mathbb{Z} \oplus \mathbb{Z}_2^2$	2	–	Native marked
14	$\mathbb{Z}$	$\mathbb{Z}_2$	$\mathbb{Z}_2^2$	$\mathbb{Z}_2$	$\mathbb{Z} \oplus \mathbb{Z}_2^4$	1	–	Native marked
15	$\mathbb{Z}$	$\mathbb{Z}_2$	$\mathbb{Z}_2^3$	$\mathbb{Z}_2$	$\mathbb{Z} \oplus \mathbb{Z}_2^5$	2	–	Native marked
16	0	0	$\mathbb{Z}_2^7$	$\mathbb{Z}_2^5$	$\mathbb{Z}_2^{12}$	–	4	Native marked
17	0	$\mathbb{Z}_2^3$	$\mathbb{Z}_2^4$	$\mathbb{Z}_2$	$\mathbb{Z}_2^8$	–	2	Native marked
18	0	$\mathbb{Z}_2^2$	$\mathbb{Z}_2^2$	$\mathbb{Z}_2$	$\mathbb{Z}_2^5$	–	2	Native marked
19	0	$\mathbb{Z}_2^2$	$\mathbb{Z}_2$	0	$\mathbb{Z}_2^3$	–	–	Native marked
20	0	$\mathbb{Z}_2^2$	$\mathbb{Z}_2^2$	$\mathbb{Z}_2$	$\mathbb{Z}_2^5$	–	2	Native marked
21	0	$\mathbb{Z}_2$	$\mathbb{Z}_2^4$	$\mathbb{Z}_2^3$	$\mathbb{Z}_2^8$	–	4	Native marked
22	0	$\mathbb{Z}_2$	$\mathbb{Z}_2^3$	$\mathbb{Z}_2^4$	$\mathbb{Z}_2^6 \oplus \mathbb{Z}_4$	–	4	Native marked
23	0	0	$\mathbb{Z}_2^4$	$\mathbb{Z}_2^2$	$\mathbb{Z}_2^6$	–	4	Native marked
24	0	$\mathbb{Z}_2^2$	$\mathbb{Z}_2^3$	$\mathbb{Z}_2$	$\mathbb{Z}_2^5$	–	2	Native marked
25	0	$\mathbb{Z}_2^4$	$\mathbb{Z}_2^8$	$\mathbb{Z}_2^9$	$\mathbb{Z}_2^{21}$	–	–	Native marked
26	0	$\mathbb{Z}_2^4$	$\mathbb{Z}_2^4$	$\mathbb{Z}_2^3$	$\mathbb{Z}_2^{11}$	–	–	Native marked
27	0	$\mathbb{Z}_2^4$	$\mathbb{Z}_2^4$	$\mathbb{Z}_2$	$\mathbb{Z}_2^9$	–	–	Native marked
28	$\mathbb{Z}_2$	$\mathbb{Z}_2^5$	$\mathbb{Z}_2^5$	$\mathbb{Z}_2^2$	$\mathbb{Z}_2^9 \oplus \mathbb{Z}_{16}$	–	–	Abstract only
29	$\mathbb{Z}_2$	$\mathbb{Z}_2^3$	$\mathbb{Z}_2$	0	$\mathbb{Z}_2^5$	–	–	Native marked
30	$\mathbb{Z}_2$	$\mathbb{Z}_2^3$	$\mathbb{Z}_2^2$	$\mathbb{Z}_2$	$\mathbb{Z}_2^5 \oplus \mathbb{Z}_4$	–	–	Abstract only
31	$\mathbb{Z}_2$	$\mathbb{Z}_2^4$	$\mathbb{Z}_2^3$	$\mathbb{Z}_2^2$	$\mathbb{Z}_2^6 \oplus \mathbb{Z}_{16}$	–	–	Abstract only
32	$\mathbb{Z}_2$	$\mathbb{Z}_2^3$	$\mathbb{Z}_2^2$	$\mathbb{Z}_2$	$\mathbb{Z}_2^5 \oplus \mathbb{Z}_4$	–	–	Abstract only
33	$\mathbb{Z}_2$	$\mathbb{Z}_2^2$	$\mathbb{Z}_2$	0	$\mathbb{Z}_2^4$	–	–	Native marked
34	$\mathbb{Z}_2$	$\mathbb{Z}_2^3$	$\mathbb{Z}_2^2$	$\mathbb{Z}_2$	$\mathbb{Z}_2^5 \oplus \mathbb{Z}_4$	–	–	Abstract only
35	0	$\mathbb{Z}_2^3$	$\mathbb{Z}_2^5$	$\mathbb{Z}_2^5$	$\mathbb{Z}_2^{13}$	–	–	Native marked
36	0	$\mathbb{Z}_2^3$	$\mathbb{Z}_2^2$	$\mathbb{Z}_2^2$	$\mathbb{Z}_2^7$	–	–	Native marked
37	0	$\mathbb{Z}_2^3$	$\mathbb{Z}_2^3$	$\mathbb{Z}_2$	$\mathbb{Z}_2^7$	–	–	Native marked
38	0	$\mathbb{Z}_2^4$	$\mathbb{Z}_2^4$	$\mathbb{Z}_2^6$	$\mathbb{Z}_2^{14}$	–	–	Native marked
39	0	$\mathbb{Z}_2^4$	$\mathbb{Z}_2^4$	$\mathbb{Z}_2^2$	$\mathbb{Z}_2^{10}$	–	–	Native marked
40	$\mathbb{Z}_2$	$\mathbb{Z}_2^4$	$\mathbb{Z}_2^4$	$\mathbb{Z}_2^2$	$\mathbb{Z}_2^7 \oplus \mathbb{Z}_{16}$	–	–	Abstract only
41	$\mathbb{Z}_2$	$\mathbb{Z}_2^3$	$\mathbb{Z}_2$	$\mathbb{Z}_2$	$\mathbb{Z}_2^4 \oplus \mathbb{Z}_4$	–	–	Abstract only
42	0	$\mathbb{Z}_2^4$	$\mathbb{Z}_2^3$	$\mathbb{Z}_2^2$	$\mathbb{Z}_2^{11}$	–	–	Native marked
43	$\mathbb{Z}_2$	$\mathbb{Z}_2^2$	$\mathbb{Z}_2$	$\mathbb{Z}_2$	$\mathbb{Z}_2^3 \oplus \mathbb{Z}_4$	–	–	Abstract only

SG	$p + ip$	MC	CF	Boson	Full group / family	Free index	H0 order	Evidence
44	0	$\mathbb{Z}_2^2$	$\mathbb{Z}_2^5$	$\mathbb{Z}_2^4$	$\mathbb{Z}_2^{11}$	—	—	Native marked
45	0	$\mathbb{Z}_2^3$	$\mathbb{Z}_2^2$	$\mathbb{Z}_2$	$\mathbb{Z}_2^6$	—	—	Native marked
46	0	$\mathbb{Z}_2^3$	$\mathbb{Z}_2^3$	$\mathbb{Z}_2^2$	$\mathbb{Z}_2^8$	—	—	Native marked
47	0	0	$\mathbb{Z}_2^8$	$\mathbb{Z}_2^{25}$	$\mathbb{Z}_2^{33}$	—	—	Native marked
48	0	$\mathbb{Z}_2$	$\mathbb{Z}_2^5$	$\mathbb{Z}_2^3$	$\mathbb{Z}_2^7 \oplus \mathbb{Z}_4$	—	—	Native marked
49	0	$\mathbb{Z}_2$	$\mathbb{Z}_2^8$	$\mathbb{Z}_2^6$	$\mathbb{Z}_2^{12} \oplus \mathbb{Z}_8$	—	—	Native marked
50	0	$\mathbb{Z}_2$	$\mathbb{Z}_2^5$	$\mathbb{Z}_2^3$	$\mathbb{Z}_2^7 \oplus \mathbb{Z}_4$	—	—	Native marked
51	0	$\mathbb{Z}_2^2$	$\mathbb{Z}_2^7$	$\mathbb{Z}_2^8$	$\mathbb{Z}_2^{17}$	—	—	Native marked
52	0	$\mathbb{Z}_2^2$	$\mathbb{Z}_2^3$	$\mathbb{Z}_2$	$\mathbb{Z}_2^6$	—	—	Native marked
53	0	$\mathbb{Z}_2^2$	$\mathbb{Z}_2^5$	$\mathbb{Z}_2^4$	$\mathbb{Z}_2^8 \oplus \mathbb{Z}_8$	—	—	Native marked
54	0	$\mathbb{Z}_2^3$	$\mathbb{Z}_2^4$	$\mathbb{Z}_2$	$\mathbb{Z}_2^8$	—	—	Native marked
55	0	$\mathbb{Z}_2^2$	$\mathbb{Z}_2^3$	$\mathbb{Z}_2^5$	$\mathbb{Z}_2^{10}$	—	—	Native marked
56	0	$\mathbb{Z}_2^2$	$\mathbb{Z}_2^3$	$\mathbb{Z}_2$	$\mathbb{Z}_2^6$	—	—	Native marked
57	0	$\mathbb{Z}_2^3$	$\mathbb{Z}_2^4$	$\mathbb{Z}_2^2$	$\mathbb{Z}_2^9$	—	—	Native marked
58	0	$\mathbb{Z}_2$	$\mathbb{Z}_2^5$	$\mathbb{Z}_2^3$	$\mathbb{Z}_2^6 \oplus \mathbb{Z}_8$	—	—	Native marked
59	0	$\mathbb{Z}_2^2$	$\mathbb{Z}_2^5$	$\mathbb{Z}_2^5$	$\mathbb{Z}_2^{12}$	—	—	Native marked
60	0	$\mathbb{Z}_2^2$	$\mathbb{Z}_2^2$	$\mathbb{Z}_2$	$\mathbb{Z}_2^5$	—	—	Native marked
61	0	$\mathbb{Z}_2^2$	$\mathbb{Z}_2$	$\mathbb{Z}_2$	$\mathbb{Z}_2^4$	—	—	Native marked
62	0	$\mathbb{Z}_2^2$	$\mathbb{Z}_2^3$	$\mathbb{Z}_2^2$	$\mathbb{Z}_2^7$	—	—	Native marked
63	0	$\mathbb{Z}_2^2$	$\mathbb{Z}_2^4$	$\mathbb{Z}_2^5$	$\mathbb{Z}_2^{11}$	—	—	Native marked
64	0	$\mathbb{Z}_2^2$	$\mathbb{Z}_2^3$	$\mathbb{Z}_2^3$	$\mathbb{Z}_2^8$	—	—	Native marked
65	0	0	$\mathbb{Z}_2^6$	$\mathbb{Z}_2^{14}$	$\mathbb{Z}_2^{20}$	—	—	Native marked
66	0	$\mathbb{Z}_2$	$\mathbb{Z}_2^6$	$\mathbb{Z}_2^5$	$\mathbb{Z}_2^9 \oplus \mathbb{Z}_8$	—	—	Native marked
67	0	$\mathbb{Z}_2$	$\mathbb{Z}_2^7$	$\mathbb{Z}_2^7$	$\mathbb{Z}_2^{15}$	—	—	Native marked
68	0	$\mathbb{Z}_2^2$	$\mathbb{Z}_2^4$	$\mathbb{Z}_2^2$	$\mathbb{Z}_2^6 \oplus \mathbb{Z}_4$	—	—	Native marked
69	0	0	$\mathbb{Z}_2^6$	$\mathbb{Z}_2^9$	$\mathbb{Z}_2^{15}$	—	—	Native marked
70	0	$\mathbb{Z}_2$	$\mathbb{Z}_2^3$	$\mathbb{Z}_2^2$	$\mathbb{Z}_2^4 \oplus \mathbb{Z}_4$	—	—	Native marked
71	0	0	$\mathbb{Z}_2^4$	$\mathbb{Z}_2^{13}$	$\mathbb{Z}_2^{17}$	—	—	Native marked
72	0	$\mathbb{Z}_2$	$\mathbb{Z}_2^4$	$\mathbb{Z}_2^3$	$\mathbb{Z}_2^9$	—	—	Native marked
73	0	$\mathbb{Z}_2^3$	$\mathbb{Z}_2^4$	$\mathbb{Z}_2$	$\mathbb{Z}_2^8$	—	—	Native marked
74	0	$\mathbb{Z}_2$	$\mathbb{Z}_2^5$	$\mathbb{Z}_2^6$	$\mathbb{Z}_2^{12}$	—	—	Native marked
75	$\mathbb{Z}$	$\mathbb{Z}_2^2$	$\mathbb{Z}_2^3$	$\mathbb{Z}_2 \oplus \mathbb{Z}_4$	$\mathbb{Z} \oplus \mathbb{Z}_2^6 \oplus \mathbb{Z}_4$	2	2	Native marked
76	$\mathbb{Z}$	$\mathbb{Z}_2^2$	$\mathbb{Z}_2$	0	$\mathbb{Z} \oplus \mathbb{Z}_2^3$	1	—	Native marked
77	$\mathbb{Z}$	$\mathbb{Z}_2^2$	$\mathbb{Z}_2^3$	$\mathbb{Z}_2$	$\mathbb{Z} \oplus \mathbb{Z}_2^6$	2	2	Native marked
78	$\mathbb{Z}$	$\mathbb{Z}_2^2$	$\mathbb{Z}_2$	0	$\mathbb{Z} \oplus \mathbb{Z}_2^3$	1	—	Native marked
79	$\mathbb{Z}$	$\mathbb{Z}_2$	$\mathbb{Z}_2^2$	$\mathbb{Z}_4$	$\mathbb{Z} \oplus \mathbb{Z}_2^3 \oplus \mathbb{Z}_4$	2	2	Native marked
80	$\mathbb{Z}$	$\mathbb{Z}_2$	$\mathbb{Z}_2$	$\mathbb{Z}_2$	$\mathbb{Z} \oplus \mathbb{Z}_2^3$	1	2	Native marked
81	$\mathbb{Z}$	$\mathbb{Z}_2^3$	$\mathbb{Z}_2^3$	$\mathbb{Z}_2^3 \oplus \mathbb{Z}_4$	$\mathbb{Z} \oplus \mathbb{Z}_2^5 \oplus \mathbb{Z}_4^3$	2	—	Native marked
82	$\mathbb{Z}$	$\mathbb{Z}_2^2$	$\mathbb{Z}_2^2$	$\mathbb{Z}_2^3 \oplus \mathbb{Z}_4$	$\mathbb{Z} \oplus \mathbb{Z}_2^3 \oplus \mathbb{Z}_4^3$	2	—	Native marked
83	$\mathbb{Z}$	$\mathbb{Z}_2$	$\mathbb{Z}_2^6$	$\mathbb{Z}_2^5 \oplus \mathbb{Z}_4$	$\mathbb{Z} \oplus \mathbb{Z}_2^3 \oplus \mathbb{Z}_4^2 \oplus \mathbb{Z}_8$	2	—	Native marked
84	$\mathbb{Z}$	$\mathbb{Z}_2$	$\mathbb{Z}_2^5$	$\mathbb{Z}_2^5$	$\mathbb{Z} \oplus \mathbb{Z}_2^8 \oplus \mathbb{Z}_8$	2	—	Native marked
85	$\mathbb{Z}$	$\mathbb{Z}_2^2$	$\mathbb{Z}_2^3$	$\mathbb{Z}_2^2 \oplus \mathbb{Z}_4$	$\mathbb{Z} \oplus \mathbb{Z}_2^5 \oplus \mathbb{Z}_4^2$	2	—	Native marked
86	$\mathbb{Z}$	$\mathbb{Z}_2^2$	$\mathbb{Z}_2^3$	$\mathbb{Z}_2 \oplus \mathbb{Z}_4$	$\mathbb{Z} \oplus \mathbb{Z}_2^4 \oplus \mathbb{Z}_4^2$	2	—	Native marked
87	$\mathbb{Z}$	$\mathbb{Z}_2$	$\mathbb{Z}_2^4$	$\mathbb{Z}_2^4 \oplus \mathbb{Z}_4$	$\mathbb{Z} \oplus \mathbb{Z}_2^6 \oplus \mathbb{Z}_4 \oplus \mathbb{Z}_8$	2	—	Native marked
88	$\mathbb{Z}$	$\mathbb{Z}_2$	$\mathbb{Z}_2^2$	$\mathbb{Z}_2 \oplus \mathbb{Z}_4$	$\mathbb{Z} \oplus \mathbb{Z}_2^2 \oplus \mathbb{Z}_4^2$	2	—	Native marked
89	0	0	$\mathbb{Z}_2^8$	$\mathbb{Z}_2^3$	$\mathbb{Z}_2^{12}$	—	4	Native marked
90	0	$\mathbb{Z}_2$	$\mathbb{Z}_2^2$	$\mathbb{Z}_2^3$	$\mathbb{Z}_2^6$	—	4	Native marked
91	0	$\mathbb{Z}_2^2$	$\mathbb{Z}_2^3$	$\mathbb{Z}_2$	$\mathbb{Z}_2^6$	—	2	Native marked

SG	$p + ip$	MC	CF	Boson	Full group / family	Free index	H0 order	Evidence
92	0	$\mathbb{Z}_2$	$\mathbb{Z}_2$	$\mathbb{Z}_2$	$\mathbb{Z}_2^3$	—	2	Native marked
93	0	0	$\mathbb{Z}_2^5$	$\mathbb{Z}_2^4$	$\mathbb{Z}_2^9$	—	4	Native marked
94	0	$\mathbb{Z}_2$	$\mathbb{Z}_2^2$	$\mathbb{Z}_2^2$	$\mathbb{Z}_2^5$	—	4	Native marked
95	0	$\mathbb{Z}_2^2$	$\mathbb{Z}_2^3$	$\mathbb{Z}_2$	$\mathbb{Z}_2^6$	—	2	Native marked
96	0	$\mathbb{Z}_2$	$\mathbb{Z}_2$	$\mathbb{Z}_2$	$\mathbb{Z}_2^3$	—	2	Native marked
97	0	0	$\mathbb{Z}_2^4$	$\mathbb{Z}_2^3$	$\mathbb{Z}_2^7$	—	4	Native marked
98	0	$\mathbb{Z}_2$	$\mathbb{Z}_2^2$	$\mathbb{Z}_2^2$	$\mathbb{Z}_2^5$	—	4	Native marked
99	0	$\mathbb{Z}_2^3$	$\mathbb{Z}_2^7$	$\mathbb{Z}_2^8$	$\mathbb{Z}_2^{18}$	—	—	Native marked
100	0	$\mathbb{Z}_2^2$	$\mathbb{Z}_2^3$	$\mathbb{Z}_2^4$	$\mathbb{Z}_2^9$	—	—	Native marked
101	0	$\mathbb{Z}_2^3$	$\mathbb{Z}_2^3$	$\mathbb{Z}_2^4$	$\mathbb{Z}_2^{10}$	—	—	Native marked
102	0	$\mathbb{Z}_2^2$	$\mathbb{Z}_2^3$	$\mathbb{Z}_2^3$	$\mathbb{Z}_2^8$	—	—	Native marked
103	0	$\mathbb{Z}_2^3$	$\mathbb{Z}_2^3$	$\mathbb{Z}_2^2$	$\mathbb{Z}_2^8$	—	—	Native marked
104	0	$\mathbb{Z}_2^2$	$\mathbb{Z}_2^2$	$\mathbb{Z}_4$	$\mathbb{Z}_2^4 \oplus \mathbb{Z}_4$	—	—	Native marked
105	0	$\mathbb{Z}_2^3$	$\mathbb{Z}_2^4$	$\mathbb{Z}_2^6$	$\mathbb{Z}_2^{13}$	—	—	Native marked
106	0	$\mathbb{Z}_2^2$	$\mathbb{Z}_2^2$	$\mathbb{Z}_2$	$\mathbb{Z}_2^5$	—	—	Native marked
107	0	$\mathbb{Z}_2^3$	$\mathbb{Z}_2^3$	$\mathbb{Z}_2^6$	$\mathbb{Z}_2^{12}$	—	—	Native marked
108	0	$\mathbb{Z}_2^3$	$\mathbb{Z}_2^2$	$\mathbb{Z}_2^4$	$\mathbb{Z}_2^9$	—	—	Native marked
109	0	$\mathbb{Z}_2$	$\mathbb{Z}_2^3$	$\mathbb{Z}_2^2$	$\mathbb{Z}_2^6$	—	—	Native marked
110	0	$\mathbb{Z}_2^2$	$\mathbb{Z}_2$	$\mathbb{Z}_2$	$\mathbb{Z}_2^4$	—	—	Native marked
111	0	0	$\mathbb{Z}_2^8$	$\mathbb{Z}_2^6$	$\mathbb{Z}_2^{14}$	—	—	Native marked
112	0	$\mathbb{Z}_2$	$\mathbb{Z}_2^5$	$\mathbb{Z}_2^4$	$\mathbb{Z}_2^8 \oplus \mathbb{Z}_4$	—	—	Native marked
113	0	$\mathbb{Z}_2^2$	$\mathbb{Z}_2^3$	$\mathbb{Z}_2^5$	$\mathbb{Z}_2^8 \oplus \mathbb{Z}_4$	—	—	Native marked
114	0	$\mathbb{Z}_2^2$	$\mathbb{Z}_2^2$	$\mathbb{Z}_2 \oplus \mathbb{Z}_4$	$\mathbb{Z}_2^3 \oplus \mathbb{Z}_4^2$	—	—	Native marked
115	0	$\mathbb{Z}_2$	$\mathbb{Z}_2^8$	$\mathbb{Z}_2^7$	$\mathbb{Z}_2^{16}$	—	—	Native marked
116	0	$\mathbb{Z}_2^2$	$\mathbb{Z}_2^4$	$\mathbb{Z}_2^3$	$\mathbb{Z}_2^7 \oplus \mathbb{Z}_4$	—	—	Native marked
117	0	$\mathbb{Z}_2^2$	$\mathbb{Z}_2^3$	$\mathbb{Z}_2^4$	$\mathbb{Z}_2^5 \oplus \mathbb{Z}_4^2$	—	—	Native marked
118	0	$\mathbb{Z}_2^2$	$\mathbb{Z}_2^3$	$\mathbb{Z}_2^4$	$\mathbb{Z}_2^5 \oplus \mathbb{Z}_4^2$	—	—	Native marked
119	0	0	$\mathbb{Z}_2^6$	$\mathbb{Z}_2^6$	$\mathbb{Z}_2^{12}$	—	—	Native marked
120	0	$\mathbb{Z}_2^2$	$\mathbb{Z}_2^3$	$\mathbb{Z}_2^4$	$\mathbb{Z}_2^5 \oplus \mathbb{Z}_4^2$	—	—	Native marked
121	0	$\mathbb{Z}_2$	$\mathbb{Z}_2^4$	$\mathbb{Z}_2^5$	$\mathbb{Z}_2^{10}$	—	—	Native marked
122	0	$\mathbb{Z}_2^2$	$\mathbb{Z}_2^2$	$\mathbb{Z}_2 \oplus \mathbb{Z}_4$	$\mathbb{Z}_2^3 \oplus \mathbb{Z}_4^2$	—	—	Native marked
123	0	0	$\mathbb{Z}_2^8$	$\mathbb{Z}_2^{19}$	$\mathbb{Z}_2^{27}$	—	—	Native marked
124	0	$\mathbb{Z}_2$	$\mathbb{Z}_2^7$	$\mathbb{Z}_2^6$	$\mathbb{Z}_2^{11} \oplus \mathbb{Z}_8$	—	—	Native marked
125	0	0	$\mathbb{Z}_2^8$	$\mathbb{Z}_2^5$	$\mathbb{Z}_2^{13}$	—	—	Native marked
126	0	$\mathbb{Z}_2$	$\mathbb{Z}_2^5$	$\mathbb{Z}_2^3$	$\mathbb{Z}_2^7 \oplus \mathbb{Z}_4$	—	—	Native marked
127	0	0	$\mathbb{Z}_2^4$	$\mathbb{Z}_2^{10}$	$\mathbb{Z}_2^{14}$	—	—	Native marked
128	0	$\mathbb{Z}_2$	$\mathbb{Z}_2^4$	$\mathbb{Z}_2 \oplus \mathbb{Z}_4$	$\mathbb{Z}_2^6 \oplus \mathbb{Z}_4 \oplus \mathbb{Z}_8$	—	—	Native marked
129	0	$\mathbb{Z}_2$	$\mathbb{Z}_2^7$	$\mathbb{Z}_2^7$	$\mathbb{Z}_2^{15}$	—	—	Native marked
130	0	$\mathbb{Z}_2^2$	$\mathbb{Z}_2^3$	$\mathbb{Z}_2^3$	$\mathbb{Z}_2^6 \oplus \mathbb{Z}_4$	—	—	Native marked
131	0	0	$\mathbb{Z}_2^6$	$\mathbb{Z}_2^{14}$	$\mathbb{Z}_2^{20}$	—	—	Native marked
132	0	0	$\mathbb{Z}_2^6$	$\mathbb{Z}_2^9$	$\mathbb{Z}_2^{15}$	—	—	Native marked
133	0	$\mathbb{Z}_2$	$\mathbb{Z}_2^4$	$\mathbb{Z}_2^3$	$\mathbb{Z}_2^6 \oplus \mathbb{Z}_4$	—	—	Native marked
134	0	0	$\mathbb{Z}_2^6$	$\mathbb{Z}_2^5$	$\mathbb{Z}_2^{11}$	—	—	Native marked
135	0	$\mathbb{Z}_2$	$\mathbb{Z}_2^3$	$\mathbb{Z}_2^4$	$\mathbb{Z}_2^8$	—	—	Native marked
136	0	0	$\mathbb{Z}_2^3$	$\mathbb{Z}_2^8$	$\mathbb{Z}_2^{11}$	—	—	Native marked
137	0	$\mathbb{Z}_2$	$\mathbb{Z}_2^4$	$\mathbb{Z}_2^5$	$\mathbb{Z}_2^{10}$	—	—	Native marked
138	0	$\mathbb{Z}_2$	$\mathbb{Z}_2^4$	$\mathbb{Z}_2^5$	$\mathbb{Z}_2^{10}$	—	—	Native marked
139	0	0	$\mathbb{Z}_2^6$	$\mathbb{Z}_2^{11}$	$\mathbb{Z}_2^{17}$	—	—	Native marked

SG	$p + ip$	MC	CF	Boson	Full group / family	Free index	H0 order	Evidence
140	0	0	$\mathbb{Z}_2^6$	$\mathbb{Z}_2^7$	$\mathbb{Z}_2^{13}$	—	—	Native marked
141	0	0	$\mathbb{Z}_2^4$	$\mathbb{Z}_2^5$	$\mathbb{Z}_2^9$	—	—	Native marked
142	0	$\mathbb{Z}_2^2$	$\mathbb{Z}_2^3$	$\mathbb{Z}_2^2$	$\mathbb{Z}_2^5 \oplus \mathbb{Z}_4$	—	—	Native marked
143	$\mathbb{Z}$	$\mathbb{Z}_2$	$\mathbb{Z}_2$	$\mathbb{Z}_3$	$\mathbb{Z} \oplus \mathbb{Z}_3 \oplus \mathbb{Z}_6^2$	1	—	Native marked
144	$\mathbb{Z}$	$\mathbb{Z}_2$	$\mathbb{Z}_2$	0	$\mathbb{Z} \oplus \mathbb{Z}_2^2$	1	—	Native marked
145	$\mathbb{Z}$	$\mathbb{Z}_2$	$\mathbb{Z}_2$	0	$\mathbb{Z} \oplus \mathbb{Z}_2^2$	1	—	Native marked
146	$\mathbb{Z}$	$\mathbb{Z}_2$	$\mathbb{Z}_2$	$\mathbb{Z}_3$	$\mathbb{Z} \oplus \mathbb{Z}_2 \oplus \mathbb{Z}_6$	1	—	Native marked
147	$\mathbb{Z}$	0	$\mathbb{Z}_2^2$	$\mathbb{Z}_3 \oplus \mathbb{Z}_6$	$\mathbb{Z} \oplus \mathbb{Z}_2 \oplus \mathbb{Z}_6^2$	2	—	Native marked
148	$\mathbb{Z}$	0	$\mathbb{Z}_2^2$	$\mathbb{Z}_6$	$\mathbb{Z} \oplus \mathbb{Z}_2^2 \oplus \mathbb{Z}_6$	2	—	Native marked
149	0	$\mathbb{Z}_2$	$\mathbb{Z}_2^2$	$\mathbb{Z}_2$	$\mathbb{Z}_2^4$	—	2	Native marked
150	0	$\mathbb{Z}_2$	$\mathbb{Z}_2^2$	$\mathbb{Z}_6$	$\mathbb{Z}_2^3 \oplus \mathbb{Z}_6$	—	2	Native marked
151	0	$\mathbb{Z}_2$	$\mathbb{Z}_2^2$	$\mathbb{Z}_2$	$\mathbb{Z}_2^4$	—	2	Native marked
152	0	$\mathbb{Z}_2$	$\mathbb{Z}_2^2$	$\mathbb{Z}_2$	$\mathbb{Z}_2^4$	—	2	Native marked
153	0	$\mathbb{Z}_2$	$\mathbb{Z}_2^2$	$\mathbb{Z}_2$	$\mathbb{Z}_2^4$	—	2	Native marked
154	0	$\mathbb{Z}_2$	$\mathbb{Z}_2^2$	$\mathbb{Z}_2$	$\mathbb{Z}_2^4$	—	2	Native marked
155	0	$\mathbb{Z}_2$	$\mathbb{Z}_2^2$	$\mathbb{Z}_2$	$\mathbb{Z}_2^4$	—	2	Native marked
156	$\mathbb{Z}_2$	$\mathbb{Z}_2^3$	$\mathbb{Z}_2^3$	$\mathbb{Z}_2^2$	$\mathbb{Z}_2^5 \oplus \mathbb{Z}_{16}$	—	—	Abstract only
157	$\mathbb{Z}_2$	$\mathbb{Z}_2^3$	$\mathbb{Z}_2^3$	$\mathbb{Z}_2 \oplus \mathbb{Z}_6$	$\mathbb{Z}_2^5 \oplus \mathbb{Z}_{48}$	—	—	Abstract only
158	$\mathbb{Z}_2$	$\mathbb{Z}_2$	$\mathbb{Z}_2$	0	$\mathbb{Z}_2^3$	—	—	Native marked
159	$\mathbb{Z}_2$	$\mathbb{Z}_2$	$\mathbb{Z}_2$	$\mathbb{Z}_3$	$\mathbb{Z}_2^2 \oplus \mathbb{Z}_6$	—	—	Native marked
160	$\mathbb{Z}_2$	$\mathbb{Z}_2^3$	$\mathbb{Z}_2^3$	$\mathbb{Z}_2^2$	$\mathbb{Z}_2^5 \oplus \mathbb{Z}_{16}$	—	—	Abstract only
161	$\mathbb{Z}_2$	$\mathbb{Z}_2$	$\mathbb{Z}_2$	0	$\mathbb{Z}_2^3$	—	—	Native marked
162	0	$\mathbb{Z}_2$	$\mathbb{Z}_2^4$	$\mathbb{Z}_2^2$	$\mathbb{Z}_2^6 \oplus \mathbb{Z}_8$	—	—	Native marked
163	0	$\mathbb{Z}_2$	$\mathbb{Z}_2^2$	$\mathbb{Z}_2$	$\mathbb{Z}_2^4$	—	—	Native marked
164	0	$\mathbb{Z}_2$	$\mathbb{Z}_2^4$	$\mathbb{Z}_2^2$	$\mathbb{Z}_2^6 \oplus \mathbb{Z}_8$	—	—	Native marked
165	0	$\mathbb{Z}_2$	$\mathbb{Z}_2^2$	$\mathbb{Z}_2$	$\mathbb{Z}_2^4$	—	—	Native marked
166	0	$\mathbb{Z}_2$	$\mathbb{Z}_2^4$	$\mathbb{Z}_2^2$	$\mathbb{Z}_2^6 \oplus \mathbb{Z}_8$	—	—	Native marked
167	0	$\mathbb{Z}_2$	$\mathbb{Z}_2^2$	$\mathbb{Z}_2$	$\mathbb{Z}_2^4$	—	—	Native marked
168	$\mathbb{Z}$	$\mathbb{Z}_2$	$\mathbb{Z}_2^2$	$\mathbb{Z}_3 \oplus \mathbb{Z}_6$	$\mathbb{Z} \oplus \mathbb{Z}_2^2 \oplus \mathbb{Z}_6^2$	2	2	Native marked
169	$\mathbb{Z}$	$\mathbb{Z}_2$	$\mathbb{Z}_2$	0	$\mathbb{Z} \oplus \mathbb{Z}_2^2$	1	—	Native marked
170	$\mathbb{Z}$	$\mathbb{Z}_2$	$\mathbb{Z}_2$	0	$\mathbb{Z} \oplus \mathbb{Z}_2^2$	1	—	Native marked
171	$\mathbb{Z}$	$\mathbb{Z}_2$	$\mathbb{Z}_2^2$	$\mathbb{Z}_2$	$\mathbb{Z} \oplus \mathbb{Z}_2^4$	2	2	Native marked
172	$\mathbb{Z}$	$\mathbb{Z}_2$	$\mathbb{Z}_2^2$	$\mathbb{Z}_2$	$\mathbb{Z} \oplus \mathbb{Z}_2^4$	2	2	Native marked
173	$\mathbb{Z}$	$\mathbb{Z}_2$	$\mathbb{Z}_2$	$\mathbb{Z}_3$	$\mathbb{Z} \oplus \mathbb{Z}_2^6$	1	—	Native marked
174	$\mathbb{Z} \oplus \mathbb{Z}_2$	$\mathbb{Z}_2^2$	$\mathbb{Z}_2^4$	$\mathbb{Z}_6^3$	$\mathbb{Z} \oplus \mathbb{Z}_2^2 \oplus \mathbb{Z}_6 \oplus \mathbb{Z}_{24} \oplus \mathbb{Z}_{48}$	1	—	Abstract only
175	$\mathbb{Z}$	$\mathbb{Z}_2$	$\mathbb{Z}_2^4$	$\mathbb{Z}_3 \oplus \mathbb{Z}_6^2$	$\mathbb{Z} \oplus \mathbb{Z}_2^5 \oplus \mathbb{Z}_{12} \oplus \mathbb{Z}_{24}$	2	—	Native marked
176	$\mathbb{Z}$	$\mathbb{Z}_2$	$\mathbb{Z}_2^3$	$\mathbb{Z}_6^2$	$\mathbb{Z} \oplus \mathbb{Z}_2^2 \oplus \mathbb{Z}_6 \oplus \mathbb{Z}_{24}$	1	—	Native marked
177	0	0	$\mathbb{Z}_2^3$	$\mathbb{Z}_2^3$	$\mathbb{Z}_2^6$	—	4	Native marked
178	0	$\mathbb{Z}_2$	$\mathbb{Z}_2^2$	$\mathbb{Z}_2$	$\mathbb{Z}_2^4$	—	2	Native marked
179	0	$\mathbb{Z}_2$	$\mathbb{Z}_2^2$	$\mathbb{Z}_2$	$\mathbb{Z}_2^4$	—	2	Native marked
180	0	0	$\mathbb{Z}_2^3$	$\mathbb{Z}_2^3$	$\mathbb{Z}_2^6$	—	4	Native marked
181	0	0	$\mathbb{Z}_2^3$	$\mathbb{Z}_2^3$	$\mathbb{Z}_2^6$	—	4	Native marked
182	0	$\mathbb{Z}_2$	$\mathbb{Z}_2^2$	$\mathbb{Z}_2$	$\mathbb{Z}_2^4$	—	2	Native marked
183	0	$\mathbb{Z}_2^2$	$\mathbb{Z}_2^4$	$\mathbb{Z}_2^5$	$\mathbb{Z}_2^{11}$	—	—	Native marked
184	0	$\mathbb{Z}_2^2$	$\mathbb{Z}_2^2$	$\mathbb{Z}_2$	$\mathbb{Z}_2^5$	—	—	Native marked
185	0	$\mathbb{Z}_2^2$	$\mathbb{Z}_2^2$	$\mathbb{Z}_2^2$	$\mathbb{Z}_2^5$	—	—	Native marked
186	0	$\mathbb{Z}_2^2$	$\mathbb{Z}_2^2$	$\mathbb{Z}_2^2$	$\mathbb{Z}_2^6$	—	—	Native marked
187	0	$\mathbb{Z}_2^2$	$\mathbb{Z}_2^3$	$\mathbb{Z}_2^2$	$\mathbb{Z}_2^{11}$	—	—	Native marked

SG	$p + ip$	MC	CF	Boson	Full group / family	Free index	H0 order	Evidence
188	$\mathbb{Z}_2$	$\mathbb{Z}_2^2$	$\mathbb{Z}_2^3$	$\mathbb{Z}_2^2$	$\mathbb{Z}_2^4 \oplus \mathbb{Z}_{16}$	—	—	Abstract only
189	0	$\mathbb{Z}_2^2$	$\mathbb{Z}_2^3$	$\mathbb{Z}_2^5 \oplus \mathbb{Z}_6$	$\mathbb{Z}_2^{10} \oplus \mathbb{Z}_6$	—	—	Native marked
190	$\mathbb{Z}_2$	$\mathbb{Z}_2^2$	$\mathbb{Z}_2^3$	$\mathbb{Z}_2 \oplus \mathbb{Z}_6$	$\mathbb{Z}_2^4 \oplus \mathbb{Z}_{48}$	—	—	Abstract only
191	0	0	$\mathbb{Z}_2^4$	$\mathbb{Z}_2^{13}$	$\mathbb{Z}_2^{17}$	—	—	Native marked
192	0	$\mathbb{Z}_2$	$\mathbb{Z}_2^4$	$\mathbb{Z}_2^4$	$\mathbb{Z}_2^5 \oplus \mathbb{Z}_8$	—	—	Native marked
193	0	$\mathbb{Z}_2$	$\mathbb{Z}_2^3$	$\mathbb{Z}_2^5$	$\mathbb{Z}_2^9$	—	—	Native marked
194	0	$\mathbb{Z}_2$	$\mathbb{Z}_2^3$	$\mathbb{Z}_2^5$	$\mathbb{Z}_2^9$	—	—	Native marked
195	0	0	$\mathbb{Z}_2^3$	$\mathbb{Z}_6$	$\mathbb{Z}_2^3 \oplus \mathbb{Z}_6$	—	4	Native marked
196	0	$\mathbb{Z}_2$	$\mathbb{Z}_2^3$	$\mathbb{Z}_3$	$\mathbb{Z}_2^2 \oplus \mathbb{Z}_{12}$	—	4	Native marked
197	0	0	$\mathbb{Z}_2^2$	$\mathbb{Z}_3$	$\mathbb{Z}_2 \oplus \mathbb{Z}_6$	—	4	Native marked
198	0	0	$\mathbb{Z}_2$	$\mathbb{Z}_3$	$\mathbb{Z}_6$	—	—	Native marked
199	0	0	$\mathbb{Z}_2$	$\mathbb{Z}_6$	$\mathbb{Z}_2 \oplus \mathbb{Z}_6$	—	2	Native marked
200	0	0	$\mathbb{Z}_2^4$	$\mathbb{Z}_2^8 \oplus \mathbb{Z}_6$	$\mathbb{Z}_2^{12} \oplus \mathbb{Z}_6$	—	—	Native marked
201	0	$\mathbb{Z}_2$	$\mathbb{Z}_2^3$	$\mathbb{Z}_6$	$\mathbb{Z}_2^3 \oplus \mathbb{Z}_{12}$	—	—	Native marked
202	0	0	$\mathbb{Z}_2^4$	$\mathbb{Z}_2^2 \oplus \mathbb{Z}_6$	$\mathbb{Z}_2^6 \oplus \mathbb{Z}_6$	—	—	Native marked
203	0	$\mathbb{Z}_2$	$\mathbb{Z}_2^3$	$\mathbb{Z}_3$	$\mathbb{Z}_2^2 \oplus \mathbb{Z}_{12}$	—	—	Native marked
204	0	0	$\mathbb{Z}_2^2$	$\mathbb{Z}_2^4 \oplus \mathbb{Z}_6$	$\mathbb{Z}_2^5 \oplus \mathbb{Z}_6$	—	—	Native marked
205	0	0	$\mathbb{Z}_2$	$\mathbb{Z}_6$	$\mathbb{Z}_2 \oplus \mathbb{Z}_6$	—	—	Native marked
206	0	$\mathbb{Z}_2$	$\mathbb{Z}_2^2$	$\mathbb{Z}_6$	$\mathbb{Z}_2^3 \oplus \mathbb{Z}_6$	—	—	Native marked
207	0	0	$\mathbb{Z}_2^6$	$\mathbb{Z}_2^2$	$\mathbb{Z}_2^8$	—	4	Native marked
208	0	0	$\mathbb{Z}_2^3$	$\mathbb{Z}_2^2$	$\mathbb{Z}_2^5$	—	4	Native marked
209	0	0	$\mathbb{Z}_2^4$	$\mathbb{Z}_2$	$\mathbb{Z}_2^5$	—	4	Native marked
210	0	$\mathbb{Z}_2$	$\mathbb{Z}_2^2$	0	$\mathbb{Z}_2^3$	—	4	Native marked
211	0	0	$\mathbb{Z}_2^3$	$\mathbb{Z}_2^2$	$\mathbb{Z}_2^5$	—	4	Native marked
212	0	0	$\mathbb{Z}_2$	$\mathbb{Z}_2$	$\mathbb{Z}_2^2$	—	2	Native marked
213	0	0	$\mathbb{Z}_2$	$\mathbb{Z}_2$	$\mathbb{Z}_2^2$	—	2	Native marked
214	0	0	$\mathbb{Z}_2$	$\mathbb{Z}_2^2$	$\mathbb{Z}_2^3$	—	4	Native marked
215	0	0	$\mathbb{Z}_2^6$	$\mathbb{Z}_2^4$	$\mathbb{Z}_2^{10}$	—	—	Native marked
216	0	0	$\mathbb{Z}_2^6$	$\mathbb{Z}_2^4$	$\mathbb{Z}_2^{10}$	—	—	Native marked
217	0	$\mathbb{Z}_2$	$\mathbb{Z}_2^3$	$\mathbb{Z}_2^4$	$\mathbb{Z}_2^8$	—	—	Native marked
218	0	$\mathbb{Z}_2$	$\mathbb{Z}_2^3$	$\mathbb{Z}_2^2$	$\mathbb{Z}_2^4 \oplus \mathbb{Z}_4$	—	—	Native marked
219	0	$\mathbb{Z}_2^2$	$\mathbb{Z}_2^3$	$\mathbb{Z}_2^2$	$\mathbb{Z}_2^3 \oplus \mathbb{Z}_4^2$	—	—	Native marked
220	0	$\mathbb{Z}_2$	$\mathbb{Z}_2$	$\mathbb{Z}_2 \oplus \mathbb{Z}_4$	$\mathbb{Z}_2 \oplus \mathbb{Z}_4^2$	—	—	Native marked
221	0	0	$\mathbb{Z}_2^6$	$\mathbb{Z}_2^{11}$	$\mathbb{Z}_2^{17}$	—	—	Native marked
222	0	$\mathbb{Z}_2$	$\mathbb{Z}_2^4$	$\mathbb{Z}_2^2$	$\mathbb{Z}_2^5 \oplus \mathbb{Z}_4$	—	—	Native marked
223	0	0	$\mathbb{Z}_2^4$	$\mathbb{Z}_2^6$	$\mathbb{Z}_2^{10}$	—	—	Native marked
224	0	0	$\mathbb{Z}_2^5$	$\mathbb{Z}_2^4$	$\mathbb{Z}_2^9$	—	—	Native marked
225	0	0	$\mathbb{Z}_2^5$	$\mathbb{Z}_2^8$	$\mathbb{Z}_2^{13}$	—	—	Native marked
226	0	0	$\mathbb{Z}_2^5$	$\mathbb{Z}_2^4$	$\mathbb{Z}_2^9$	—	—	Native marked
227	0	0	$\mathbb{Z}_2^4$	$\mathbb{Z}_2^4$	$\mathbb{Z}_2^8$	—	—	Native marked
228	0	$\mathbb{Z}_2^2$	$\mathbb{Z}_2^3$	$\mathbb{Z}_2$	$\mathbb{Z}_2^4 \oplus \mathbb{Z}_4$	—	—	Native marked
229	0	0	$\mathbb{Z}_2^5$	$\mathbb{Z}_2^7$	$\mathbb{Z}_2^{12}$	—	—	Native marked
230	0	$\mathbb{Z}_2$	$\mathbb{Z}_2^2$	$\mathbb{Z}_2^2$	$\mathbb{Z}_2^3 \oplus \mathbb{Z}_4$	—	—	Native marked